

# PORTFOLIO TUGAS AKHIR

**SISTEM PENGUSIR HEWAN HAMA (OTTER) PADA  
PERIKANAN POKDAKAN MINA JAYA DENGAN  
NOTIFIKASI AKTIVASI VIA TELEGRAM**



# **PEST ANIMAL REPELLENT SYSTEM (OTTER) IN MINA JAYA POKDAKAN FISHERIES WITH ACTIVATION NOTIFICATION VIA TELEGRAM**

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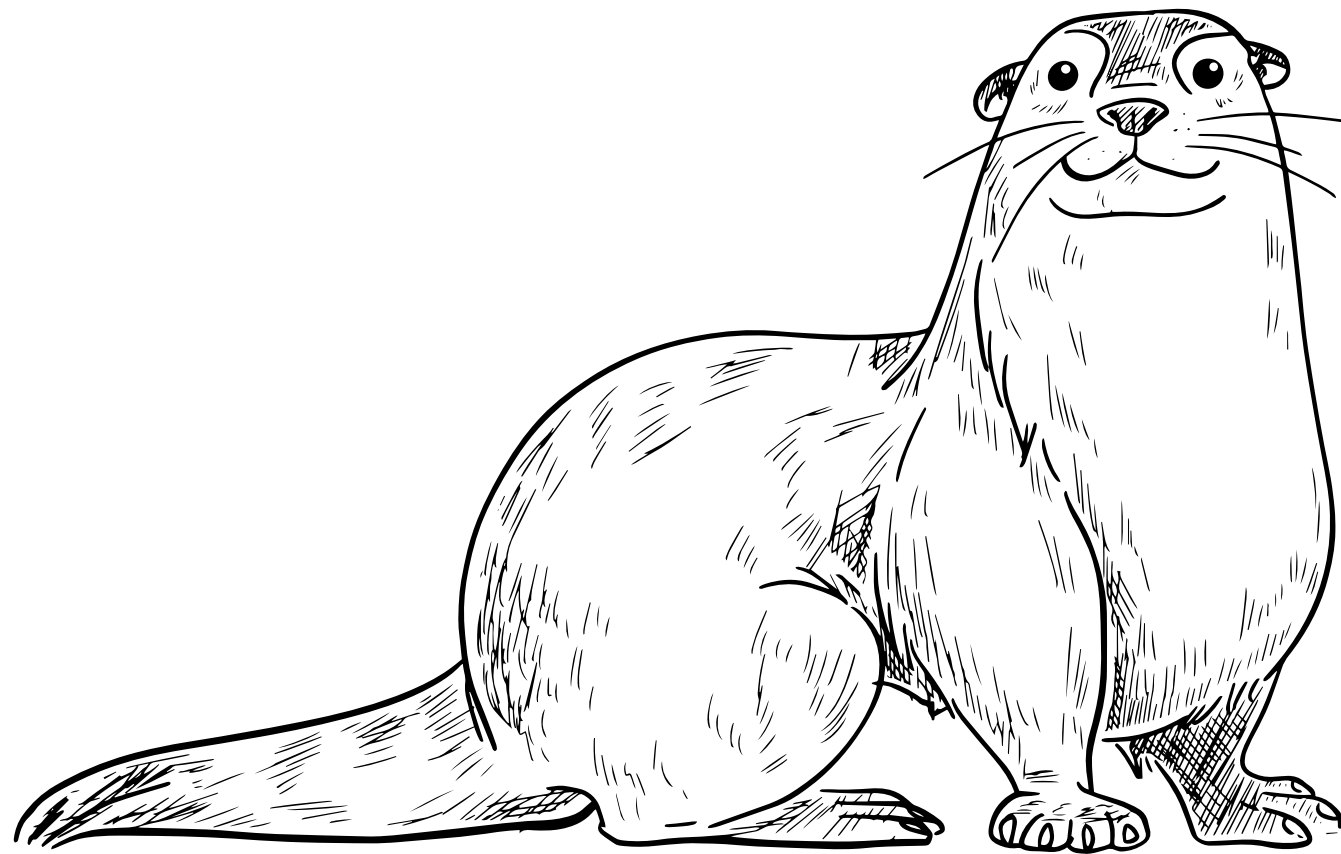
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# CHAPTER 1

## Introduction

# Background of the Problem

- Decline in income due to a decrease in the number of fish that can be harvested.
- Increase in operational costs to replace lost fish and repair damaged infrastructure.
- Decrease in the competitiveness of the aquaculture industry due to rising production costs.
- Potential decline in investment interest in the aquaculture sector.





# Formulation of the problem

**1**

**How effective is the device in detecting disturbances using a gyroscope sensor to repel weasel pests?**

**2**

**How does the system perform in response to angle changes, and is the device effective in operation?**

**3**

**What is the shape of the ultrasonic wave generated in various testing scenarios?**

**4**

**How reliable is the Telegram notification system across different testing scenarios?**

# Objectives and Benefit

## Objectives

This garangan repellent device offers a practical, efficient solution to protect fish farms from attacks. It is simple, user-friendly, and affordable, making it ideal for small to medium-scale fish farmers.

## Benefit

This tool supports investment in fish farming by offering reliable protection from external threats. It repels garangan without harming the fish, promoting sustainable aquaculture and reducing the need for constant supervision.

## **Scope of Problem**

**Battery life is only around 12 hours and cannot operate for a full 24 hours.**

**The main material is still acrylic, which is less resistant to cold weather, especially rain.**

**The device can be carried away by currents as it floats without a support pole.**

**The system cannot yet distinguish between garangan pests and other animals.**

# Research Methods

## Literature Review

Studied relevant technologies such as gyroscope sensors, ultrasonic waves, and Telegram notifications as the foundation for system design.

## System Design

Designed the system architecture, pest detection workflow, and integration of key components like ESP32, MPU-6050, and ultrasonic buzzer.

## Hardware & Implementation

Installed components on a floating platform, programmed the ESP32, and configured the system to operate automatically when detecting movement in the pond.

## Testing & Data Analysis

Conducted quantitative testing to evaluate sensor response, ultrasonic effectiveness, and Telegram notification reliability.



# **CHAPTER 2**

## **Literature Review and theorecital basis**

# Literature Review

- Annisa et al. (2020) – Prototype of an IoT-based bird repellent system with hybrid power supply.
- Mohammad Iqbal et al. (2022) – Rice field rat repellent device using Arduino Uno and ultrasonic waves.
- Khoirudin Fathoni et al. (2024) – Development of an IoT-based bird repellent and automatic soil pH monitoring tool for Mangunsari farmer group.



# Theoretical Basis

## System Using Internet of Things

The system is connected to the internet, allowing real-time monitoring and notifications

## Wireshark Network Monitoring

Wireshark is software used to analyze computer network activity, useful for network professionals, administrators, researchers, and network software developers.

## Notifications using Telegram Bot

Telegram Bot API is used for notifications, enabling real-time monitoring of router status in the network.

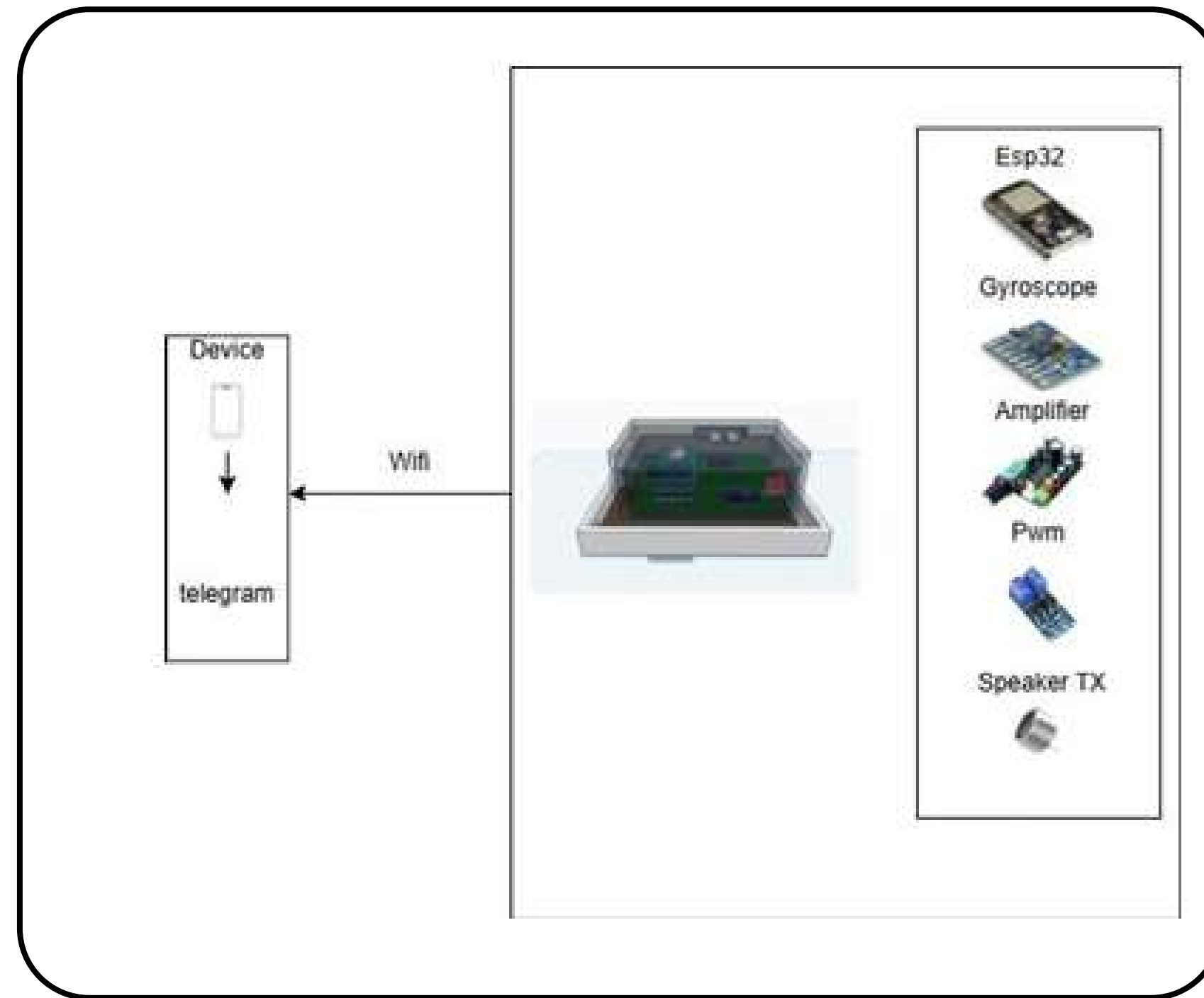


# **CHAPTER 3**

## **Research Methodology and system Design**

# System Design

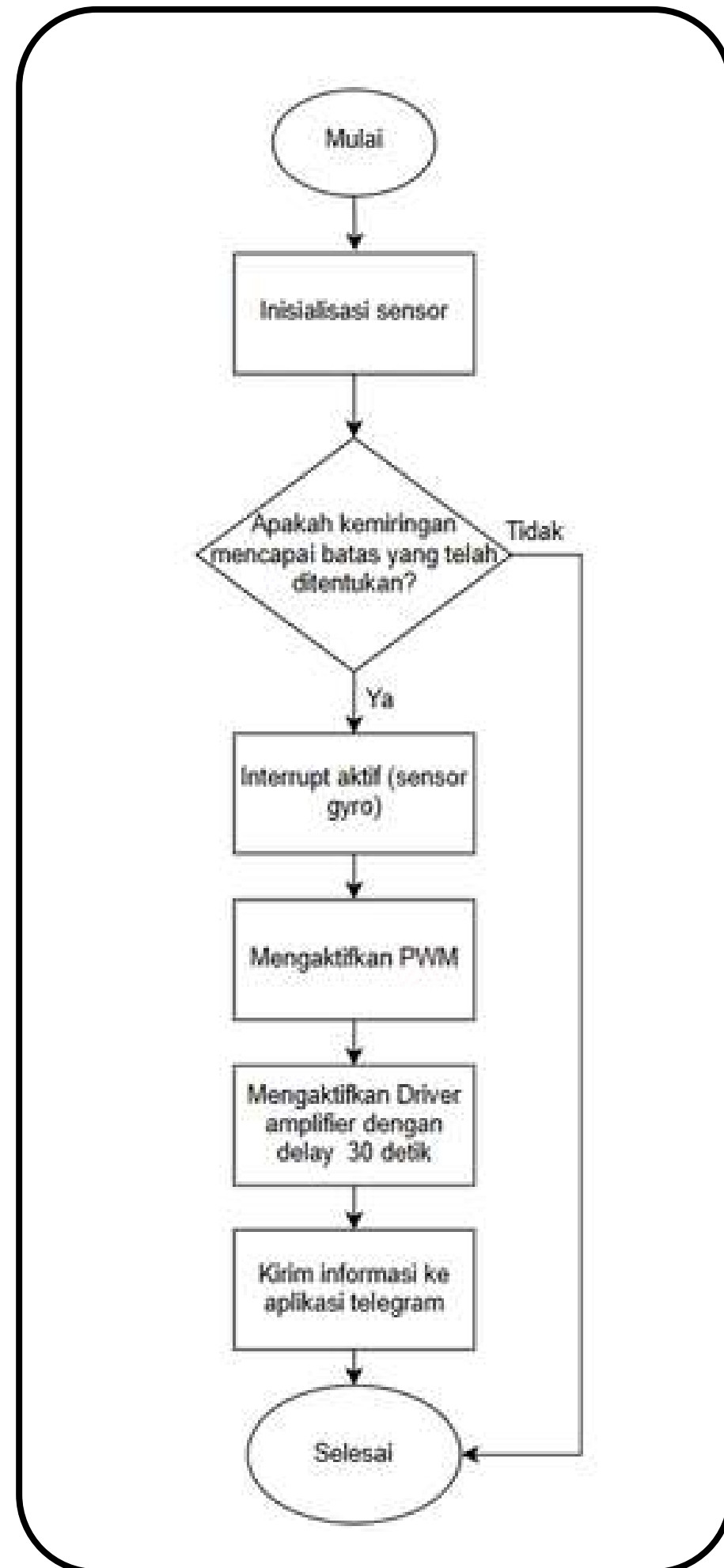
## Capstone Design



The system uses WiFi to connect the ESP32 with Telegram for automatic pest repellent monitoring. The ESP32 processes gyroscope data to detect vibrations or movement, then activates the PWM module to generate ultrasonic waves, which are amplified and emitted through a speaker. Notifications are sent to the user's phone via Telegram.



# Block Diagram



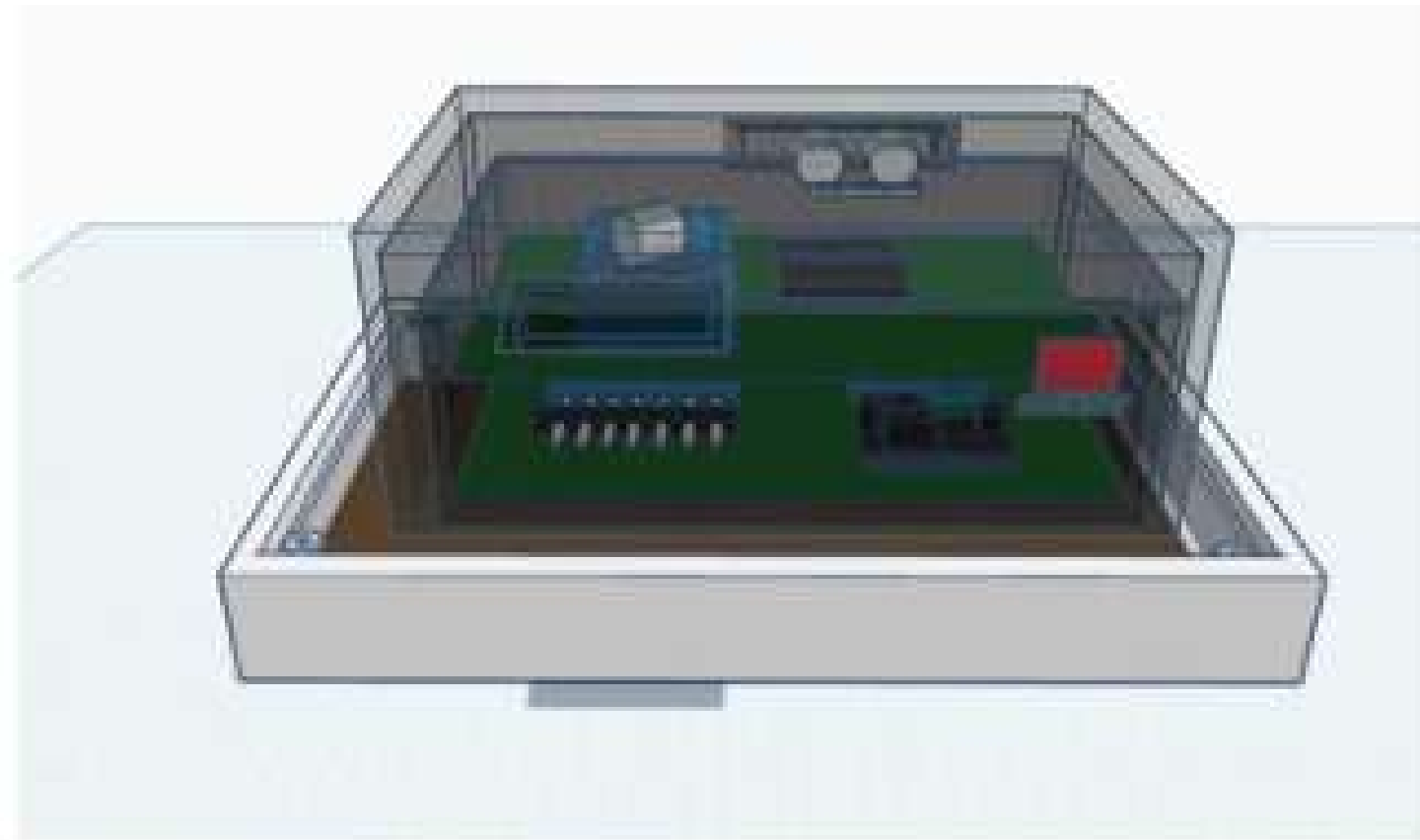
## Capstone Design

The system uses a 3-axis gyroscope sensor to detect suspicious movements around the fish pond. Data from the sensor is processed by the ESP32 to activate the PWM module, which generates ultrasonic signals. These signals are amplified and transmitted through the TX speaker to repel weasel pests.



## System Hardware Design

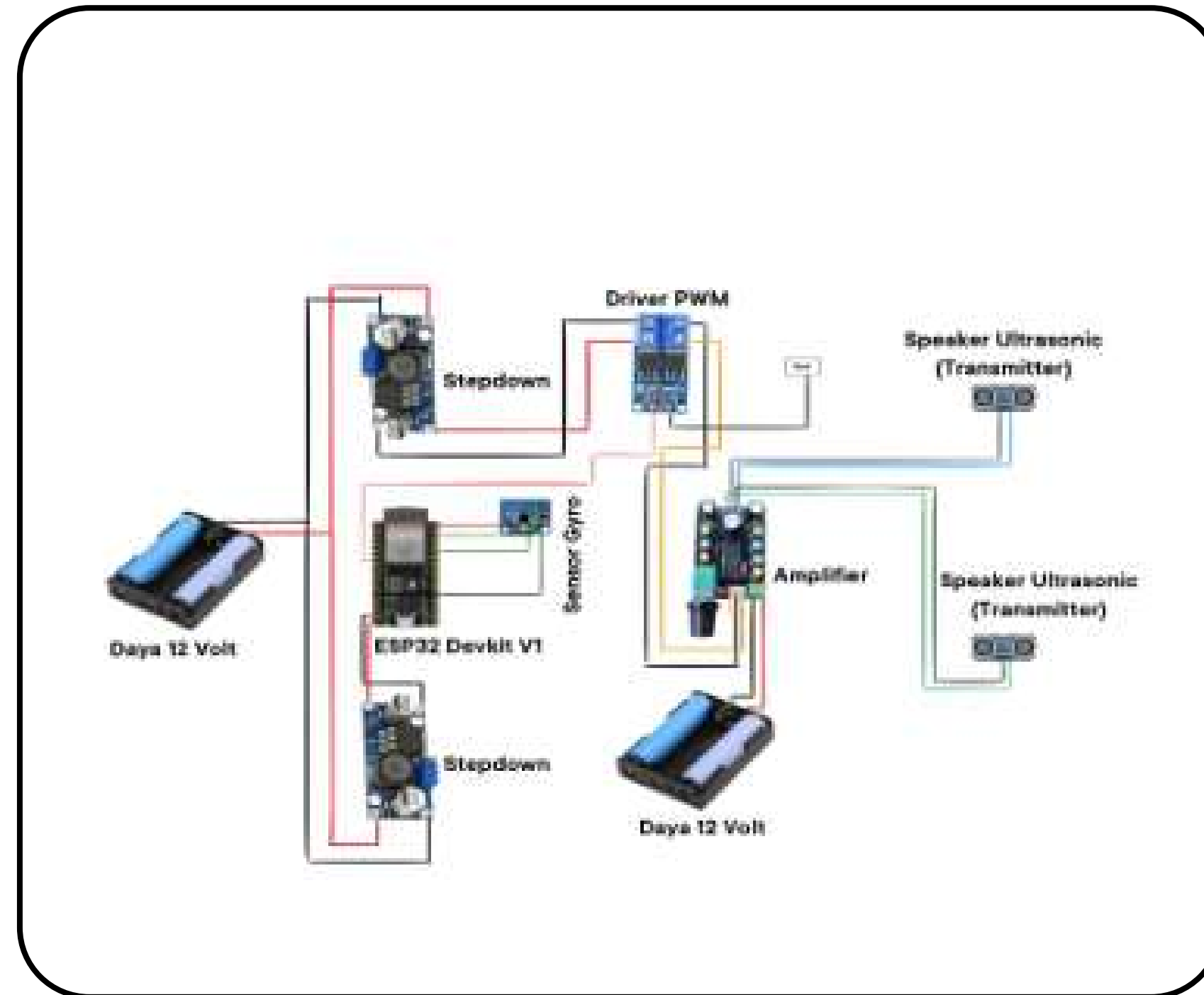
## Capstone Design



The device uses styrofoam for buoyancy, with added pipes for stability in the pond. Electronic components are centrally mounted on acrylic and arranged to maintain balance and prevent gyroscope interference. The bottom is lined with aluminum to direct ultrasonic waves downward.

# IoT Device System Design

## Capstone Design



The device consists of several key hardware components, including the ESP32 microcontroller, step-down voltage regulator, amplifier, 18650 lithium battery, and TX speaker. Each component plays an essential role in ensuring the system functions reliably and efficiently.

# CHAPTER 4

## Result Analysis

Tilt testing on gyroscope and frequency sensors **Test Result**

| Nomor Pengujian | Sudut drajat | frekuensi yang dikeluarkan | Rata rata frekuensi yang dikeluarkan            |
|-----------------|--------------|----------------------------|-------------------------------------------------|
| Pengujian 1     | 2,5          | 20666 Hz                   | <i>Rata rata frekuensi</i><br>20<br>= 21.816 Hz |
| Pengujian 2     | 3,0          | 20666 Hz                   |                                                 |
| Pengujian 3     | 3,5          | 20666 Hz                   |                                                 |
| Pengujian 4     | 4,0          | 20888 Hz                   |                                                 |
| Pengujian 5     | 4,5          | 20888 Hz                   |                                                 |
| Pengujian 6     | 5,0          | 20888 Hz                   |                                                 |
| Pengujian 7     | 5,5          | 21111 Hz                   |                                                 |
| Pengujian 8     | 6,0          | 21222 Hz                   |                                                 |
| Pengujian 9     | 6,5          | 22444 Hz                   |                                                 |
| Pengujian 10    | 7,0          | 21555 Hz                   |                                                 |
| Pengujian 11    | 7,5          | 21777 Hz                   |                                                 |
| Pengujian 12    | 8,0          | 21777 Hz                   |                                                 |
| Pengujian 13    | 8,5          | 21777 Hz                   |                                                 |
| Pengujian 14    | 9,0          | 22444 Hz                   |                                                 |
| Pengujian 15    | 9,5          | 22000 Hz                   |                                                 |
| Pengujian 16    | 10,0         | 22222 Hz                   |                                                 |
| Pengujian 17    | 10,5         | 22222 Hz                   |                                                 |
| Pengujian 18    | 11,0         | 22444 Hz                   |                                                 |
| Pengujian 19    | 11,5         | 22444 Hz                   |                                                 |
| Pengujian 20    | 12,0         | 26222 Hz                   |                                                 |

The device can float stably on water and detect slight tilts caused by waves using a gyroscope sensor. The greater the tilt, the higher the ultrasonic frequency produced averaging 21816 Hz and peaking at 26222 Hz at a 12° angle. These frequencies effectively repel pests without disturbing humans, indicating the system is responsive and adaptive to changes in water surface conditions.

Testing on wireshark  
application Test Result

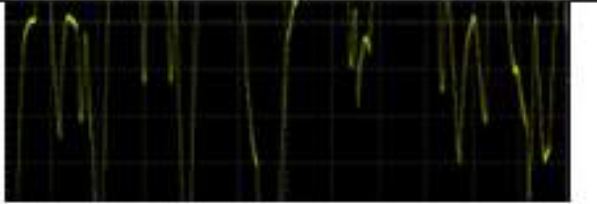
| Pengujian   | Throughput<br>(bits/s) | Packet loss<br>(%) | Rata rata<br>delay (ms) | Rata rata jitter<br>(ms) |
|-------------|------------------------|--------------------|-------------------------|--------------------------|
| Pengujian 1 | 14,09k                 | 7,8                | 79,231425               | 1,897974                 |
| Pengujian 2 | 5,989                  | 10.49              | 172653464,5             | 980.162,5222             |
| Pengujian 3 | 6.024                  | 12,99              | 247962295,9             | 14.322,09738             |
| Pengujian 4 | 15,46k                 | 6.99               | 132.235.395             | 4265684,556              |
| Pengujian 5 | 7.781k                 | 12.3               | 233.302.200             | 13.611,51407             |
| Pengujian 6 | 7253                   | 12.68              | 243.0588.56,<br>7       | 2762,176162              |
| Pengujian 7 | 5580                   | 12.37              | 235.748.236,<br>9       | -12404,18118             |
| Pengujian 8 | 13k                    | 8.4                | 200754806,7             | –<br>0,00000001623<br>33 |
| Pengujian 9 | 5280                   | 12.46              | 256158141,8             | -23046,09929             |

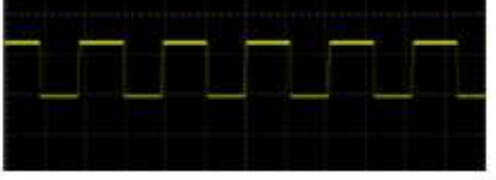
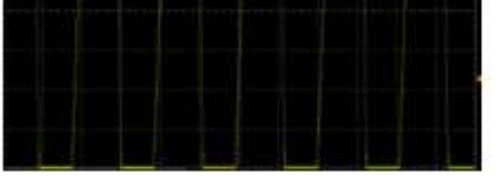
|              |       |      |                   |                          |
|--------------|-------|------|-------------------|--------------------------|
| Pengujian 10 | 9864  | 8.8  | 182.247.825,<br>3 | –<br>0,00000001499<br>75 |
| Pengujian 11 | 6129  | 8.29 | 247.812.098,<br>9 | 1.501.100,486            |
| Pengujian 12 | 5005  | 8.7  | 243544750,8       | -79516845,9              |
| Pengujian 13 | 4.956 | 8.7  | 159.455.243,<br>2 | -3431506,757             |
| Pengujian 14 | 12k   | 8.3  | 202573064,2       | -3640354,158             |
| Pengujian 15 | 7441  | 9.9  | 217.012.492       | –<br>0,00000001515<br>75 |
| Pengujian 16 | 11k   | 7,9  | 167749683,9       | 16751443,08              |
| Pengujian 17 | 9614  | 5.29 | 181056839,6       | -50063981,13             |
| Pengujian 18 | 5030  | 4.89 | 260203403         | -105600134,3             |
| Pengujian 19 | 8838  | 8.5  | 171.693.994       | -20127126,51             |
| Pengujian 20 | 5084  | 13.2 | 299170763,1       | 20946643,08              |

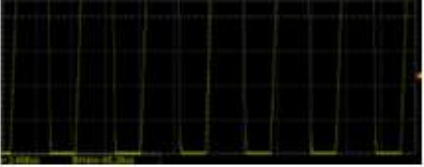
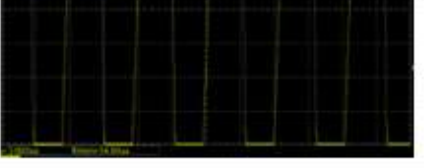
Testing showed varied network performance. Results revealed fluctuating throughput, high packet loss, extreme delays, and unstable jitter. This indicates the system still needs improvements in timing accuracy and transmission stability before field deployment.



PWM testing on an  
oscilloscope Test Result

| Pengujian   | Frekuensi (kHz) | Gelombang frekuensi                                                                  |
|-------------|-----------------|--------------------------------------------------------------------------------------|
| Pengujian 1 | 23.8            |    |
| Pengujian 2 | 34.7            |   |
| Pengujian 3 | 29.4            |  |

|             |      |                                                                                       |
|-------------|------|---------------------------------------------------------------------------------------|
| Pengujian 4 | 19.5 |    |
| Pengujian 5 | 19.7 |   |
| Pengujian 6 | 24.0 |  |
| Pengujian 7 | 24.0 |  |

|              |      |                                                                                       |
|--------------|------|---------------------------------------------------------------------------------------|
| Pengujian 8  | 26.6 |    |
| Pengujian 9  | 24.3 |   |
| Pengujian 10 | 26.0 |  |
| Pengujian 11 | 23.6 |  |

Oscilloscope results show the device produces 19.5–34.7 kHz ultrasonic signals. Waveforms ranged from smooth sine like to sharp square waves. Some instability appeared in later tests, likely due to interference or PWM issues. Overall, the device works well as we expected.



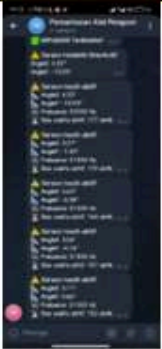
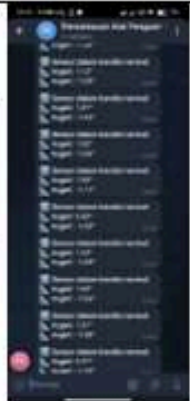
Testing notifications on telegram

Test Result

| Pengujian    | Notifikasi            | Status    | Waktu (Detik) |
|--------------|-----------------------|-----------|---------------|
| Pengujian 1  | Notif <i>ON</i>       | Berfungsi | 6             |
| Pengujian 2  | Notif <i>ON</i>       | Berfungsi | 6             |
| Pengujian 3  | Notif <i>ON</i>       | Berfungsi | 7             |
| Pengujian 4  | Notif <i>ON</i>       | Berfungsi | 6             |
| Pengujian 5  | Notif <i>OFF</i>      | Berfungsi | 4             |
| Pengujian 6  | Notif <i>OFF</i>      | Berfungsi | 5             |
| Pengujian 7  | Notif <i>OFF</i>      | Berfungsi | 4             |
| Pengujian 8  | Notif <i>OFF</i>      | Berfungsi | 4             |
| Pengujian 9  | Ubah <i>Threshold</i> | Berfungsi | 6             |
| Pengujian 10 | Ubah <i>Threshold</i> | Berfungsi | 6             |
| Pengujian 11 | Ubah <i>Threshold</i> | Berfungsi | 7             |
| Pengujian 12 | Ubah <i>Threshold</i> | Berfungsi | 6             |
| Pengujian 13 | Durasi Aktif          | Berfungsi | 3             |
| Pengujian 14 | Durasi Aktif          | Berfungsi | 3             |
| Pengujian 15 | Durasi Aktif          | Berfungsi | 3             |
| Pengujian 16 | Durasi Aktif          | Berfungsi | 3             |
| Pengujian 17 | Status Keseluruhan    | Berfungsi | 4             |
| Pengujian 18 | Status Keseluruhan    | Berfungsi | 4             |
| Pengujian 19 | Status Keseluruhan    | Berfungsi | 4             |
| Pengujian 20 | Status Keseluruhan    | Berfungsi | 4             |

Test results show the Telegram notification system respond well. The bot sends complete alerts when the tilt exceeds the threshold, with a 3–7 second response time. Features like Notif ON/OFF, Threshold Adjustment, Active Duration, and System Status work properly. Overall, the notification feature can function is stable and responsive

Battery Life Test for 12 hours Test Result

| Pengujian | Status Pengujian | Penjelasan                                                                                           | Gambar                                                                                |                                                                                       |
|-----------|------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|           |                  |                                                                                                      | Alat                                                                                  | Telegram                                                                              |
| 1.        | Berhasil         | Dilakukan pengujian alat pada ruangan terbuka pada tanggal 4 sampai 5 mei sekitar 12 jam 6 menit.    |      |     |
|           |                  |                                                                                                      |    |    |
| 2.        | Berhasil         | Dilakukan pengujian alat pada ruangan terbuka pada tanggal 11 sampai 12 mei sekitar 11 jam 32 menit. |    |    |
| 3.        | Berhasil         | Dilakukan pengujian alat pada ruangan terbuka pada tanggal 5 mei sekitar 11 jam 31 menit.            |    |    |
|           |                  |                                                                                                      |  |   |
| 4.        | Berhasil         | Dilakukan pengujian alat pada ruangan terbuka pada tanggal 12 mei sekitar 12 jam 3 menit.            |  |  |

The battery test results show that the device has an average endurance of **11-12 hours**, depending on sensor activity and data processing. This proves that the system has **good battery life** and is **suitable for night time monitoring**, especially nocturnal pests such as mongooses, when the pond is not monitored.

Weather Resistance  
Testing Test Result

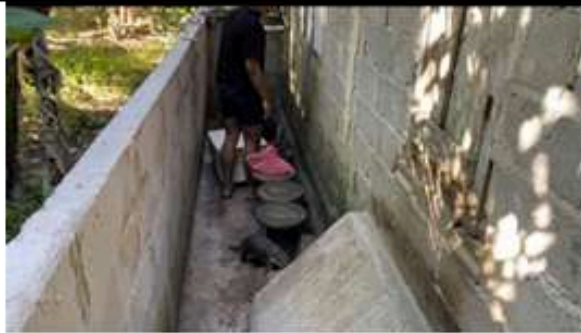
| Pengujian Pada saat Cuaca Panas |                  |                                                                                     |                                                                                                                                  |
|---------------------------------|------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Nomor Pengujian                 | Status Pengujian | Gambar                                                                              | Penjelasan                                                                                                                       |
| 1.                              | Berhasil         |   | Pengujian dilakukan pada tanggal 4 Mei saat terjadi hujan dengan intensitas cukup deras, yang berlangsung selama 1 jam 28 menit. |
|                                 |                  |  |                                                                                                                                  |
| 2.                              | Berhasil         |  | Pengujian dilakukan pada tanggal 11 Mei saat terjadi hujan dengan intensitas deras, berlangsung selama 2 jam 9 menit.            |

| Pengujian Pada saat Cuaca Panas |          |                                                                                       |                                                                                                                                                                                                                                                                                    |
|---------------------------------|----------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.                              | Berhasil |  | 4 Mei, pengujian dimulai pukul 13.13 siang. Cuaca saat itu relatif teduh, namun sempat diguyur hujan ringan sekitar pukul 13.30. Menariknya, cuaca kembali cerah dan panas mulai pukul 15.00 hingga malam hari. Ini memberikan variasi kondisi yang cukup ekstrem dalam satu hari. |

The test results show the device is weather-resistant, functioning well in both heat and heavy rain. Sensor accuracy was maintained, with minor delays likely due to signal interference. Condensation was resolved using silica gel. Overall, the device remained stable with proper precautions.



Testing on otters  
Test Result

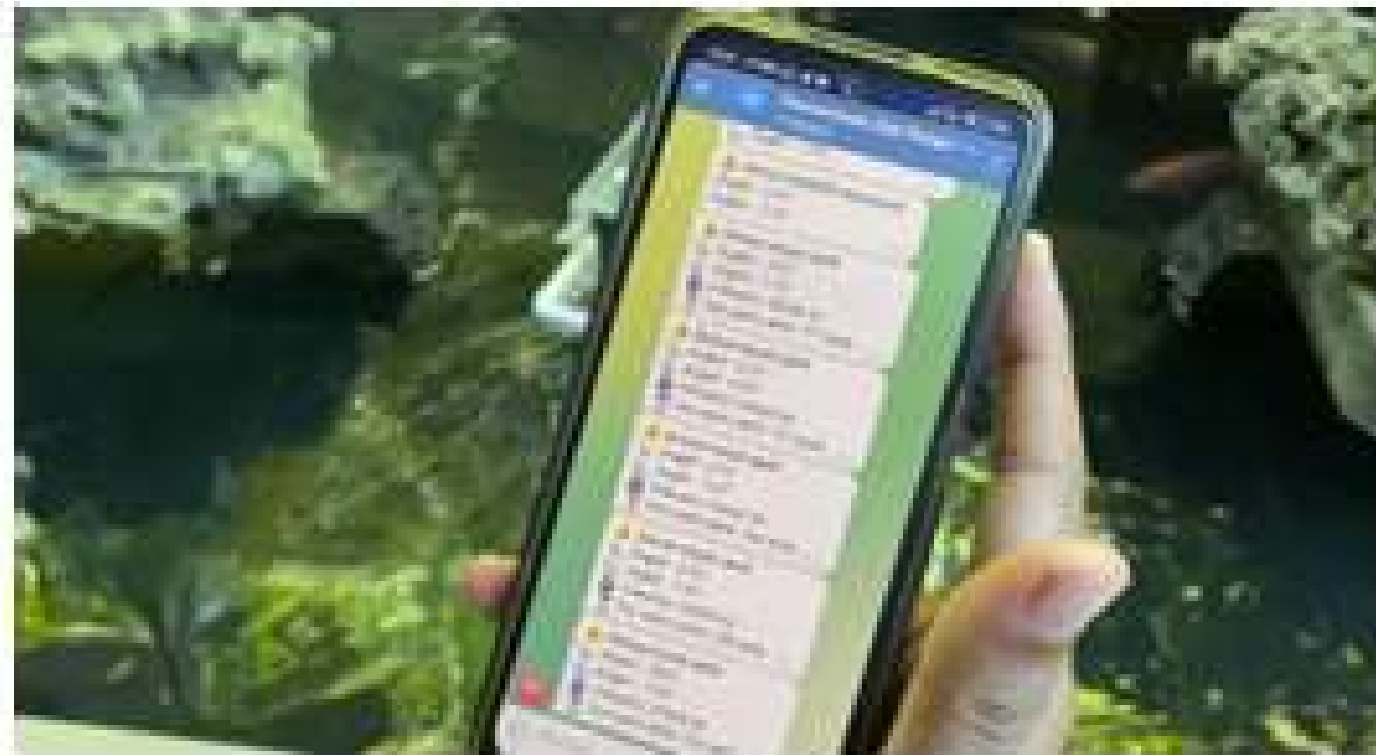
| Pengujian pada hewan |                                                                                     |                                                      |
|----------------------|-------------------------------------------------------------------------------------|------------------------------------------------------|
| Pengujian            | Gambar                                                                              | Penjelasan                                           |
| 1.                   |   | Penguji memasuki kandang otter tanpa gangguan        |
| 2.                   |  | Otter tersebut mulai mendekati kolam yang disediakan |

|    |                                                                                       |                                                                               |
|----|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 3. |    | Otter tersebut mulai berenang kedalam kolam yang tadi di sediakan             |
| 4. |   | Penguji memasukkan alat kedalam kolam yang berisi otter dan otter menjauhinya |
| 5. |  | Penguji melakukannya berulang kali untuk mengusir otter dari dalam kolam      |

|    |                                                                                       |                                                                                        |
|----|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 6. |    | Penguji belum sempat meletakkan alat otter tersebut sudah terlebih dahulu menjauhi     |
| 7. |  | Otter tersebut mengeluarkan suara yang keras pada saat alat lama diletakkan pada kolam |

The test on an otter showed consistent avoidance behavior after device activation. In the fourth trial, the otter showed stress and aggression, so testing was stopped. The results show that the device can disturb animals such as otters with a certain time delay.

## Testing in the fish pond **Test Result**



For the last experiment, the device was placed in a pond containing many fish, and the fish had no interference reaction to the device placed on the surface of the pond.



# CHAPTER 5

## CONCLUSION



## Conclusion

The device can detect vibration/movement automatically and accurately using a gyroscope sensor.

The system showed stable performance and responsiveness to angle changes, and was able to adaptively adjust the frequency according to environmental conditions.

Although the frequency remains high, the waveforms in some tests are less stable and noisy, possibly due to interference or suboptimal PWM configuration.

Telegram notifications (ON/OFF, change threshold, duration, status) performed well with a response time of 3-7 seconds, proving the system is fast and reliable.

**THANK YOU**

M A T U R N U W U N